
Encode
Shuffle
Analyze
Trusted Setup
Groth16 setup for the main proof
pk → Clients
vk → Shuffler
Enrollment Attestation Authority (EA)
federated enrollment anchors
threshold attestation
attested enrollment metadata
mobility-capable parameters
mode commitment
secret commitment
Enrollment / Warmup
of the aggregation path
warmup updates state
not aggregated
N_turn = max(K, 2)
data flow
trust / assumption
check passed
Stage 1: Client-Side Processing
1 Local trajectory state
hidden coordinates
timestamp
previous / anchor commitments
2 Payload-binder construction
derive hidden primary cell
construct public payload binders
bind the payload to the committed primary
3 Commitment-chain update
location commitments
commitment chain
binds state / round / payload context
4 Proof generation
per-step trajectory continuity
ADNC window bound
hidden primary-cell derivation
payload-binder consistency
secret / mode consistency
a single proof object
client report = proof + public binders + encrypted payload package
Stage 2: Shuffler-Side Verification
1 chain-state check
predecessor commitment, timestamp monotonicity
2 public-signal consistency
enrollment state, anchor, mode
3 ciphertext-package integrity check
transport consistency
4 verify proof
trajectory and state consistency
5 check payload binders
payload consistent with the committed primary
6 state update / hold / blacklist
replay / identity / drift
invalid updates are rejected before secure aggregation
Payload swapping is rejected if the payload binders are inconsistent or the proof fails.
Verification & Security Semantics
continuity soundness
zero knowledge
payload-binding integrity
public-output DP
The proof enforces trajectory continuity and payload-binding consistency.
Public binders tie the committed primary to the payload contents.
Transcript integrity protects the encrypted package in transit.
accepted reports only
Stage 3: Dual Secure Aggregation & DP Release
Aggregation
Aggregator A
decrypt payload
re-check format / context
add valid share
A and R do not collude
Aggregator R
decrypt payload
re-check format / context
add valid share
Decoder
reconstruct aggregate
DP Release
SVT threshold filtering
Laplace perturbation
Published DP heatmap
only accepted reports reach aggregation
public-output DP applies only to the released heatmap
A and R must not collude